

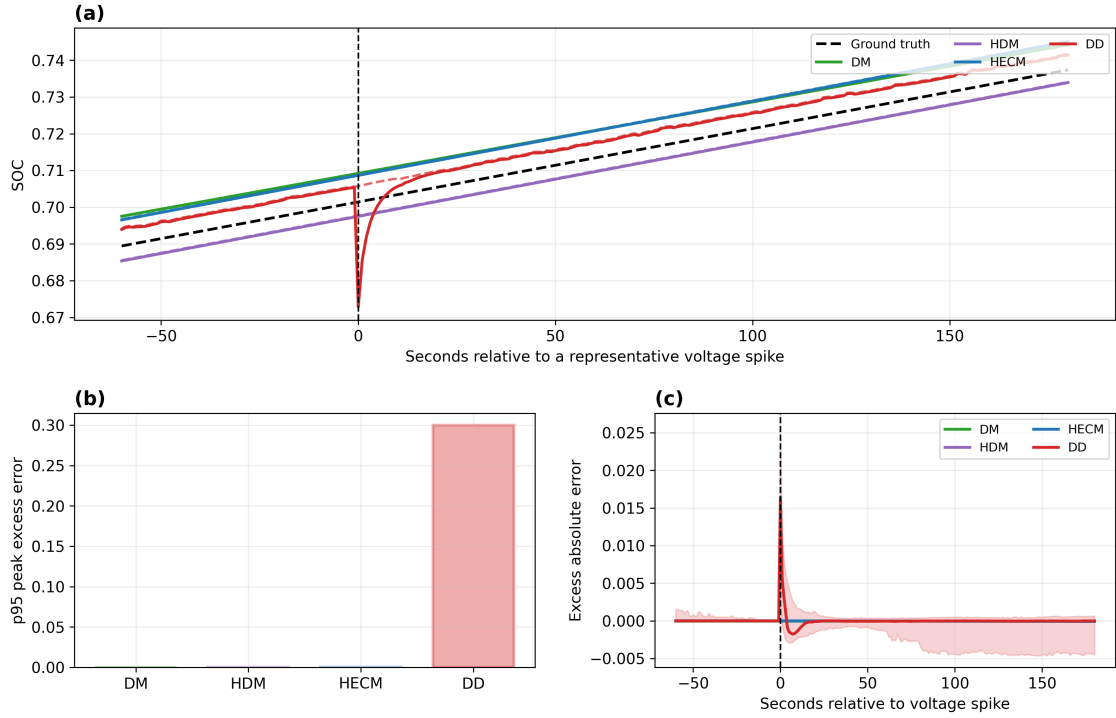


Figure 6.13: Event-aligned voltage-spike response. Panel (a) compares representative SOC trajectories with dataset SOC. Panel (b) reports the model-wise 95th percentile peak excess error. Panel (c) shows mean excess absolute error aligned across repeated spikes. The local DD response decays rapidly and is therefore weakly represented by full-window MAE.

6.4.8 ADC quantization

Figure 6.14 compares representative raw and quantized current and voltage signals and summarizes the six-cell ΔMAE . The tested steps of $\Delta I = 0,01\text{ A}$, $\Delta U = 0,005\text{ V}$, and $\Delta T = 0,5^\circ\text{C}$ preserve local signal shape but affect the representatives differently. DM and HDM increase by 0.0038 and 0.0035 MAE. HECM changes by -0.0018 and DD by $+0.0002$. Every confidence interval overlaps zero.

The small response of DM and HDM is consistent with repeated current-rounding errors entering cumulative charge. HECM combines quantized current with voltage-residual correction, while DD embeds all quantized channels in a multichannel temporal context. Their signed changes remain inside the uncertainty range and do not indicate physical improvement or exact cancellation. At the selected coarse resolution, ADC quantization produces smaller penalties than systematic current error, sample loss, and burst dropout.